

2023EBN122A01

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Electricity and Electronics EBN 122

25 July 2023

Test Information

Maximum marks:	10	Full marks:	10
Due date:	1 August 2023, 23:59	Open/Closed book:	Open
Total number of pages (this page included)	7		

Lecturers: Prof P. de Villiers, Prof. P.J Jacobs**Assistant Lecturers:** N.-J Lord, M. Malesela, G. Neeuwfan**Important**

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers, including complex numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your assessment, you declare the following:**

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

I hereby declare that the work herewith is completely my own and that no parts have been copied in any way from current or previous students or other sources.

6. Contact ebn2023.consult@gmail.com **before** the deadline for assistance with technical issues.

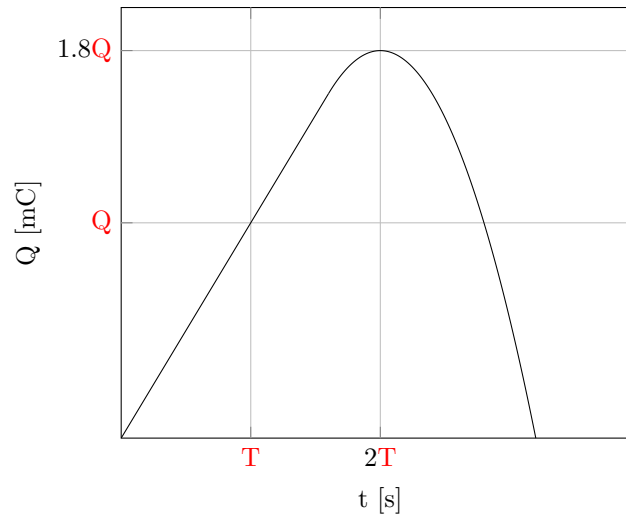
Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct:** $1-j2$ or $-3+j4$
 - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
 - Capital J is however incorrect, and $1-2j$ is also not acceptable.
 - If a complex number has no real part, for example $0+j2$, write $0+j2$ as the final answer.
- **Correct:**
 - With angle in degrees: $12\angle 34^\circ$;
 - With angle in radians: $12\angle 0.5934$
 - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect:**
 - $1-i2$ (Use j instead of i)
 - $1 - j2$ (no spaces between characters)
 - $12 \angle 34 \text{ deg}$; (no spaces between characters)
 - $j10+1$ (Real part before the imaginary part for rectangular form.)
 - $-3+4j$ (should be $j4$ instead of $4j$)

Section 1**Questions 1 to 2 [2]**Given: Q , T .

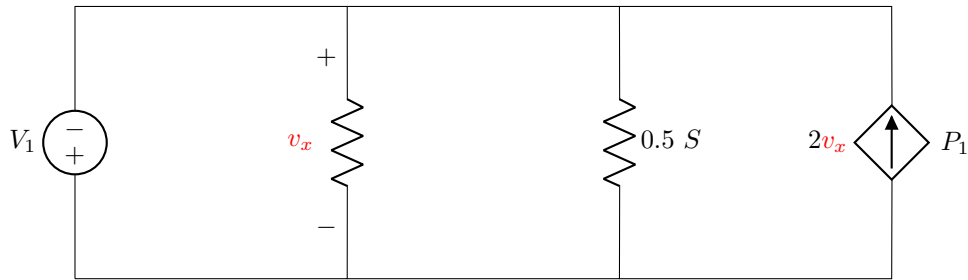
The charge flowing in a conductor is shown below.

**01** Determine the current at $t = 2T$ s [1]**02** Determine the current at $t = T$ s [1]

Question 3 [2]	
Given: Q , X , E .	
03	To move $-Q$ nC in X μ s through an element requires E mJ. Determine the voltage V across the element, assuming a constant current. [2]

Question 4 [2]	
Given: t , E , v .	
04	The total energy used by a light bulb in t hours is E kJ. The voltage across the bulb is v V. Determine the current I through the bulb. [2]

Questions 5 to 6 [2]	
Given: X , Y .	
The voltage across an element is $V(t) = \frac{di}{dt}$ V and the current through the element is $i(t) = Xe^{-Yt}$ mA. The power associated with the element as a function of time is $p(t) = Ae^{Bt}$ (in μW).	
05	Determine A [1]
06	Determine B [1]

Question 7 [2]Given: v_x .**07** Determine the power associated with P_1 [2]**Total Section 1: [10]****Grand Total : [10]**